

Bilge Kağan Pamuk

Unity Developer & Systems Engineer · XR, Simulation & Multiplayer

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Professional Summary

Unity Developer and Systems Engineer with 4+ years of experience taking VR, MR, and WebGL simulation products from architecture to store release. Shipped dental and medical training applications holding a 4.2-star average rating with 2,000+ active users on Google Play and the App Store. Specialized in multiplayer XR (Photon Fusion 2, Meta Avatar SDK), Firebase-backed real-time systems, and hardware–software co-design (Arduino, ESP32, CAN bus). MSc candidate in Health Informatics at Hacettepe University (GPA 4.00/4.00), researching XR in medical education.

Core Skills

Unity & C#: Unity 2022 / Unity 6, C#, OOP, ScriptableObjects, Addressables, prefab and scene management, Unity Profiler, Sentry

XR & Platforms: Meta Quest 2/3, HoloLens 2, Varjo, Apple Vision Pro, WebGL, iOS, Android, Windows Standalone

Networking & Backend: Photon Fusion 2, Photon Voice 2, Firebase (Auth, Firestore, Realtime DB), REST APIs, JSON, async/await, WebSocket basics

Hardware & Embedded: Arduino (Mega, Uno, Nano), ESP32, ESP32-CAM, Raspberry Pi, CAN bus, sensor integration, serial communication

AI & LLM Integration: OpenAI, Anthropic Claude, and Google Gemini APIs; prompt engineering; NMT systems

Shaders & Graphics: HLSL, Unity Shader Graph, visual effect systems for automotive HMI and simulation

Programming & Tools: C# (primary), Python, TypeScript, SQL, C++ (basic); Git, GitLab, GitHub, CI/CD, Postman, Figma, ProtoPie, Jira

Professional Experience

Unity Developer & Research Engineer — MEDEASOFT A.Ş.

Feb 2023 – Present

Ankara, Türkiye · Full-time

- Develop and ship VR and WebGL products in Unity (C#) for dental training, surgical planning, and anatomy education; products hold a 4.2-star average rating on Google Play and the App Store with 2,000+ active users
- Architect multiplayer VR applications with Photon Fusion 2, Photon Voice 2, and Meta Avatar SDK v40; led the Photon Fusion 1 to Fusion 2 migration on Meta Quest 3
- Integrate Firebase (Firestore, Auth, Realtime Database) for real-time student progress tracking, room management, and dynamic data synchronization
- Build modular, content-driven gamified learning systems with ScriptableObjects, prefab management, and Addressables
- Implement REST API integrations with asynchronous JSON data flows; own error handling and QA test checklists within the build pipeline
- Profile and optimize builds with Unity Profiler and Sentry; maintain GitLab repositories with branch-based workflows
- Lead technical documentation and presentations for one TÜBİTAK 1507 and two TÜBİTAK 1501 national R&D projects

Product Design Engineer — SIMUX Defence

Nov 2024 – Dec 2025

Cambridge, UK (remote) · Freelance contract alongside full-time role

- Designed a Unity-based automotive HMI simulator for the Aston Martin Lagonda digital cockpit; integrated CAN bus protocols for real-time vehicle telemetry
- Conducted eye-tracking studies (HP Reverb, Varjo) on driver attention patterns, information hierarchy, and interaction flows
- Built HLSL shader effects and visual feedback systems for instrument cluster displays

Mobile App Developer — Speeder System

Nov 2023 – Apr 2024

Aalsmeer, Netherlands (remote) · Freelance

- Researched drone operator workflows and designed tablet-based control interfaces in Unity, validated through iterative prototyping
- Built the product demo showcased at the Berlin Aerospace Fair, gathering live feedback from aviation professionals

Application Developer — Akgun Technology

Sep 2022 – Feb 2023

Ankara · Full-time

- Contributed to Unity-based simulation and game development projects in a cross-functional engineering team

Selected Projects

Morfo VR & Morfo Mobile — Dental Morphology Learning Platform	2023 - 2026
• Cross-platform dental morphology learning app for Meta Quest 3 (VR) and iOS/Android; gamified exercise modules built on ScriptableObjects, modular prefabs, Addressables • Photon Fusion 2 Shared Mode multiplayer with Meta Avatar SDK v40 and Photon Voice 2; Firebase backend for progress tracking and real-time sync • Companion research paper under review at the International Journal of Serious Games (IJSG)	
Digital Twin Mixed Reality System — XR + Arduino Hardware Control Panel	2026 - Ongoing
• Meta Quest 3 MR app paired with physical Arduino Uno + ESP32 control panel for industrial and healthcare simulation • MRI/CT data rendered as interactive 3D volumes; collaborative manipulation across networked headsets via Photon Fusion 2 • Bidirectional real-time data flow between physical sensors and Unity via serial communication and an ESP32 wireless bridge	
MediBot — AI-Powered Turkish Health Assistant	2026 - Ongoing
• Multi-LLM mobile health assistant (Unity client + TypeScript backend) routing across OpenAI GPT, Anthropic Claude, and Google Gemini APIs • Firebase Auth and Firestore, image-analysis pipeline, three-tier subscription system with dynamic model routing	

ECE VR Brain Surgery Simulation — Inter-University Competition	2022 - 2023
• Immersive VR brain surgery training simulation with surgical interaction mechanics, haptic-style feedback, anatomical 3D model integration	

Education

MSc, Health Informatics (English-medium) — Hacettepe University	Sep 2025 - Present
GPA 4.00/4.00 · Research: XR in medical education, LLM-based localization for Unity simulation, digital simulation effectiveness	
BSc, Software Engineering — Atılım University	2018 - 2023
GPA 3.43/4.00 · Full merit scholarship (4 years) · Focus on HCI, UX design, XR development; Erasmus exchange at University of Ostrava, Czech Republic (2022)	

Publications

1. <i>“Digital Simulation Technologies in Dental Local Anesthesia Education”</i> — Health and Technology (Springer), under review · First author 2. <i>“NMT vs. LLMs for Medical Simulation Localization in Unity”</i> — International Journal of Serious Games (IJSG), under review 3. <i>“Evaluating Usability and User Experience in a Virtual Reality-Based Driving Simulation”</i> — International Informatics and Software Engineering Conference, 2026 4. <i>“Understanding User Experience (UX) for Virtual Reality Based Surgical Training”</i> — EDULEARN Conference Proceedings (D. Topalli, Y. E. Cetin, B. K. Pamuk, et al.) 5. <i>“A Temporomandibular Joint Course with Metaverse Experience”</i> — International Eastern Conference on Human–Computer Interaction (HCI 2023), 2023	
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Languages

Turkish (native) · English (C1, professional working proficiency)